Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 16)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 Table 1 shows the textural composition of four soil samples.

Table 1

Soil sample	Sand / %	Silt / %	Clay / %
A	85	10	5
B	40	40	20
C	20	25	55
D	10	80	10

0 1 . 1 Identify the sample that is a loam and explain why loams are ideal for cultivating most crops.

[3 marks]

0 1 . 2 Explain why sample A is likely to have the highest permeability but the lowest nutrient retention.

[3 marks]

0 1 . 3 Compare the thermal capacity and root penetration of soils A and C.

[2 marks]

Do not write outside the box

Do not write outside the box

.....

.....

0 1 . 4 Describe how the texture of a soil sample could be determined by sedimentation.

[2 marks]

.....

.....

.....

.....

10

0 2 Figure 1 shows the daily deaths in London and the mean concentration of smoke in the air during the smog of December 1952.

Figure 1

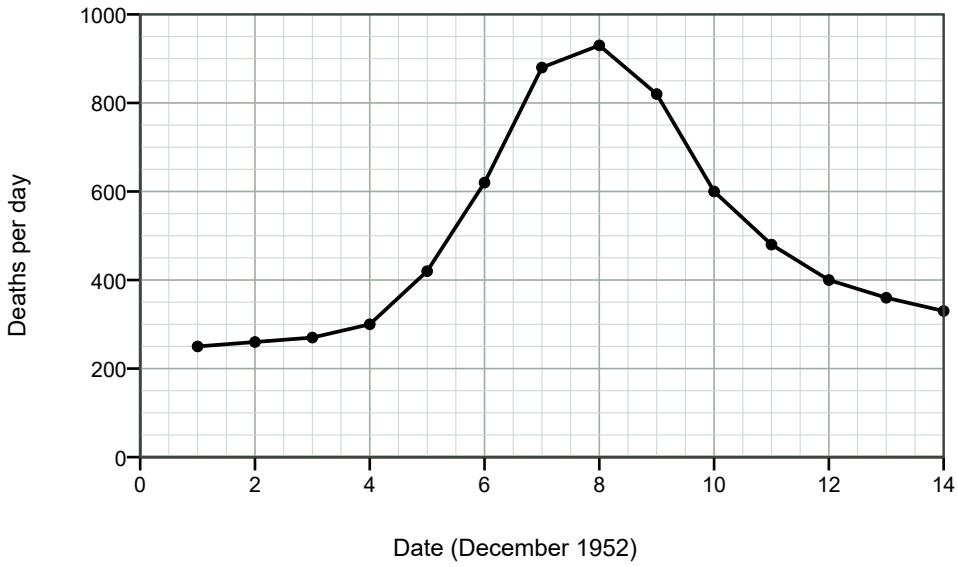


Table 1

Date	3 Dec	5 Dec	7 Dec	8 Dec	10 Dec	12 Dec
Smoke / mg m^{-3}	0.4	1.5	4.4	4.5	1.2	0.5

0 2 . 1 Describe the relationship between smoke concentration and daily deaths shown in Figure 1 and Table 1.

[2 marks]

.....

Do not write outside the box

0 2 . 2 Calculate the percentage increase in daily deaths between 1 December and 8 December.

[2 marks]

Answer ____ %

0 2 . 3 Explain how the weather conditions in December 1952 caused the smog to form.

[3 marks]

0 2 . 4 Explain why most of the deaths were among the elderly, the very young and people with existing respiratory disease.

[2 marks]

0 2 . 5 State the legislation introduced in the UK in response to this smog.

[1 mark]

.....

10

0 3 Table 1 shows the average annual radiation dose received by a person in the UK from different sources.

Table 1

Source	Percentage of average dose / %
Radon gas from the ground	50
Medical (X-rays, CT scans)	16
Gamma rays from rocks / buildings	13
Internal (food and drink)	9.5
Cosmic radiation	10
Occupational, fallout, nuclear discharges, consumer products	1.5

0 3 . 1 Explain why radon concentrations are high inside some houses in Devon and Cornwall.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 3 . 2 Use Table 1 to calculate the percentage of the average dose that comes from artificial (human-made) sources.

[1 mark]

Answer ..... %

Do not write outside the box

0 3 . 3 Suggest **two** reasons why the dose received by an individual may differ greatly from the average.

[2 marks]

1 .....

.....

2 .....

.....

0 3 . 4 Explain why a risk:benefit analysis is needed for the use of X-rays in medicine, and give **two** reasons why the risks of ionising radiation are difficult to assess.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

10

0 4 . 1 Explain **three** advantages of biofuels over other renewable energy resources.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 4 . 2 Explain **three** disadvantages of biofuel crops.

[3 marks]

Do not write outside the box

Do not write outside the box

.....

.....

.....

.....

.....

.....

.....

0 4 . 3 Describe **two** new biofuel technologies.

[2 marks]

1 .....

.....

2 .....

.....

0 4 . 4 Explain how new low-temperature turbine technology could allow geothermal electricity generation in the UK.

[2 marks]

.....

.....

.....

.....

10

0 5 . 1 Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

A High liposolubility and high persistence ☐

B Low liposolubility and low persistence ☐

C High liposolubility and low persistence ☐

D Low liposolubility and high persistence ☐

0 5 . 2 Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

.....

Do not write outside the box

0 5 . 3 Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

5

0 6 Table 1 shows three investigations carried out by a class.

Table 1

Investigation	Data collected
A	Mean height of bluebells in a coppiced and an uncoppiced wood (both normally distributed)
B	Number of limpets in each of five zones on a rocky shore, compared with the numbers expected if they were evenly spread
C	Soil moisture and the percentage cover of a moss, measured at 14 points along a transect

0 6 . 1 Name the most suitable statistical test for each investigation in Table 1.

[3 marks]

0 6 . 2 Explain what it means if a test gives a probability of $p = 0.02$.

[2 marks]

Do not write outside the box

0 6 . 3 Explain why the calculated value from a test must be compared with a critical value, and what else is needed to find that critical value.

[3 marks]

0 6 . 4 Explain what the **standard deviation** of each set of bluebell heights tells the student, and how it is shown on a graph of the means.

[3 marks]

0 6 . 5 The class found a significant correlation in investigation C. Explain **four** reasons why this does not show that soil moisture causes the moss to grow.

[4 marks]

Do not write outside the box

.....

.....

.....

.....

15

0 7 . 1 Explain how integrated manufacture saves energy in the iron and steel industry.

[2 marks]

.....

.....

.....

.....

0 7 . 2 Explain why recycling is not always energy efficient.

[2 marks]

.....

.....

.....

.....

0 7 . 3 Explain how each of the following manages electricity infrastructure to conserve energy.

[4 marks]

High-voltage grid .....

.....

.....

.....

ICT management of supply and demand .....

.....

.....

.....

0 7 . 4 Give **two** locational factors affecting where new generating infrastructure is built.

[2 marks]

1 .....

.....

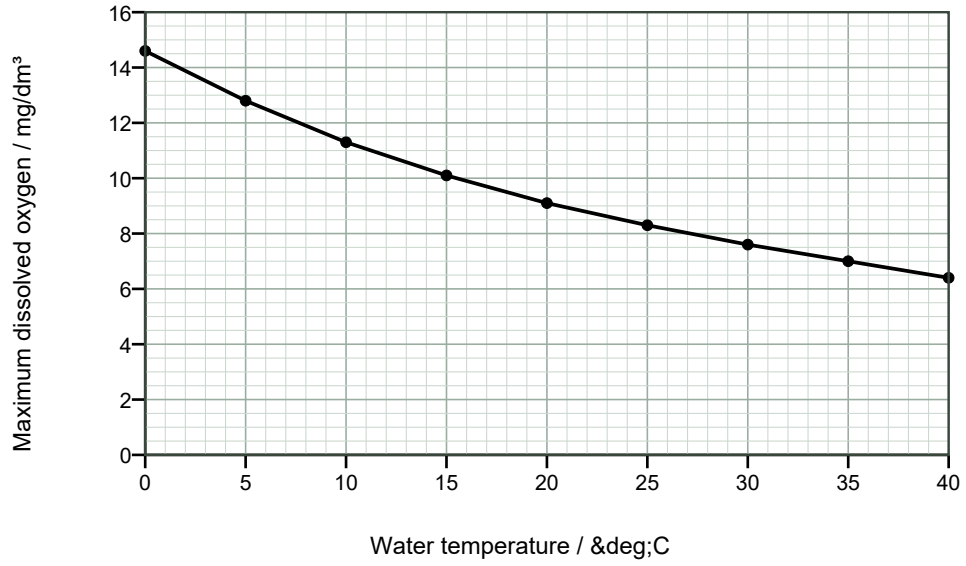
2 .....

.....

10

0 8 Figure 1 shows the maximum concentration of dissolved oxygen in fresh water at different temperatures. A power station discharges warm cooling water into a river. The lowest dissolved oxygen concentration that trout can tolerate is 6.5 mg dm^{-3}.

Figure 1



0 8 . 1 Describe the relationship shown in Figure 1.

[1 mark]

.....

.....

0 8 . 2 Use Figure 1 to determine the highest water temperature at which trout could survive if the water were fully saturated with oxygen.

[1 mark]

Do not write outside the box

Answer [ ] °C

0 8 . 3 In summer the river upstream of the power station is at 20 °C and the effluent raises the temperature by 7 °C. Use Figure 1 to calculate the reduction in maximum dissolved oxygen concentration caused by the effluent.

[2 marks]

Answer [ ] mg/dm³

0 8 . 4 Explain why the actual dissolved oxygen concentration downstream may be much lower than the values in Figure 1.

[2 marks]

.....

.....

.....

.....

0 8 . 5 Describe **three** effects of warmer water on aquatic organisms other than reduced dissolved oxygen.

[3 marks]

1 .....

.....

2 .....

.....

3 .....

.....

Do not write outside the box

0 8 . 6 Describe how a cooling tower reduces the temperature of power station effluent water.

[1 mark]

.....

.....

10

0 9 . 1 Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

A A waste is assumed to cause pollution until research shows it is unlikely to do so ☐

B The polluter must pay for any damage caused ☐

C Emissions should be as low as reasonably achievable ☐

D Only pollutants known to be toxic need to be controlled ☐

0 9 . 2 Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

.....

.....

.....

.....

0 9 . 3 Give **two** choices about emission location that reduce the severity of pollution.

[2 marks]

1 .....

.....

2 .....

.....

5

Do not write outside the box

1 0 . 1 Evaluate the success of the methods and strategies used to reduce global climate change.

[9 marks]

Do not write outside the box

Do not write outside the box

Extra space

10 . 2 State what is meant by a **carbon footprint**.

[1 mark]

10

Write an essay on one of the following topics.

11 . 1 Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.

[25 marks]

OR

11 . 2 Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

END OF QUESTIONS